Strtok-extract tokens from strings

strtok is a function in C programming that stands for "string tokenize." It's used to break a string into a sequence of tokens (substrings) based on a specified delimiter. The function takes two main arguments: the string to be tokenized and a string containing a set of delimiter characters.

```
char *strtok(char *str, const char *delim);
```

string − This is the string which is passed as the parameter which is to be modified by this function.

delimiter − This is the string containing the delimiter on the basis of which string will be broken into series of tokens.And these may vary from one call to another.

Return value

Strtok function returns a pointer to the first token found in the string. A null pointer is returned if there are no tokens left to retrieve.

Hello bye----->strtok--------->hello
.....>bye

Extracting a single token

Q:
```
#include<stdio.h>
#include<string.h>
int main()
 {
        char str[]="good evening students";
        char delim[]=" ";
        char *token;
        token=strtok(str,delim);
        printf("%s\n",token);
}
```

output: good
//

Extracting all tokens

```
#include<stdio.h>
#include<string.h>
```

```
int main()
{
	char str[30]="hello-how -are-you";
	char delim[2]="-";
	char *token;
	token=strtok(str,delim);

	while(token!=NULL)
	{
	  printf("%s\n",token);
	  token=strtok(NULL,delim);
	}

}
```
output:
hello
how
are
you

//
1.
```
#include<stdio.h>

#include<string.h>

 int main()

   {

	  char str[30]="vector india institute";

	  char *token=strtok(str," ");

	  printf("%s\n",token);

	  token=strtok(NULL," ");

	  printf("%s\n",token);

	  token=strtok(NULL," ");

	  if(token!=NULL)

	 {

			printf("still nore to search\n");

			printf("%s\n",token);

	 }

	 else

	  {

	     printf("completed\n");

	  }
```

```
          token=strtok(NULL," ");
          if(token==NULL)
            {
                  printf("no more to search/completed\n");
            }
   else
    {
        printf("%s",token);
   }
}
```

2.

//

Q:
```
#include<stdio.h>
#include<string.h>
int main()
 {
    int count=0;
    char str[]="good evening students";
    char delim[]=" ";
    char *token;
    token=strtok(str,delim);

    while(token!=NULL)
    {
          count++;
      printf("%s\n",token);
      token=strtok(NULL,delim);
    }
    printf("number of words is %d\n",count);

  }
```

output: good
evening
students
number of words is 3

//

Q:
```
#include<stdio.h>
```

```
#include<string.h>
 int main()
  {

        char str[]="hello abcd,vector;india-bye";
        char delim[]=" -,;";
        char *token;
        token=strtok(str,delim);
        while(token!=NULL)
        {
          printf("%s\n",token);
          token=strtok(NULL,delim);

        }
 }
```

output

hello
abcd
vector
india
bye

//

```
q:#include<stdio.h>
#include<string.h>
int main()
{

        char str[]="helloabcdvectorindia bye";
        char delim[]=" -,;";
        char *token;
        token=strtok(str,delim);
        while(token!=NULL)
        {
          printf("%s\n",token);
          token=strtok(NULL,delim);

        }
}
```

OUTPUT: **helloabcdvectorindia**
bye

manipulate the string

```
#include<stdio.h>
#include<string.h>
int main()
{
```

```
    char str[]="vector-india-institute";
    char delim[]="-";
    char *token;
    token=strtok(str,delim);
    printf("%s\n",token);
    printf("checking the string is %s\n",str);
}
```

output
vector

```
#include<stdio.h>
#include<string.h>
int main()
{

    char str[]="vector-india-institute";
    char delim[]="-";
    char *token;
    token=strtok(str,delim);
    printf("%s\n",token);
    printf("checking the string is %s\n",str);
    token=strtok(NULL,delim);
    printf("second token collected is %s\n",token);

    printf("again checking oringinal string is %s\n",str);
}
```

Using strtok on constant or read-only strings

```
char* s = "This is in read-only memory!";
char* tok = strtok(s, " ");
```

Segmentation fault

String Length of word

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main(void) {
   char string[] = "Hello world!";
   char* token= strtok(string, " ");

   while(token!= NULL)
   {
      printf("%d\n", strlen(token));

    token = strtok(NULL, " ");
```

```
    }
}
```

Q:longest word

```
#include<stdio.h>
#include<string.h>
int main()
{
    char str[] = "This is a samplee string";
    char *token, *longestWord = "";
    int  maxLength = 0;

    token = strtok(str, " ");
    while (token != NULL)
    {
        int  length = strlen(token);
        if (length > maxLength)
        {
            maxLength = length;
            longestWord = token;
        }
        token = strtok(NULL, " ");
    }

    printf("Longest word: %s\n", longestWord);

    return 0;
}
```